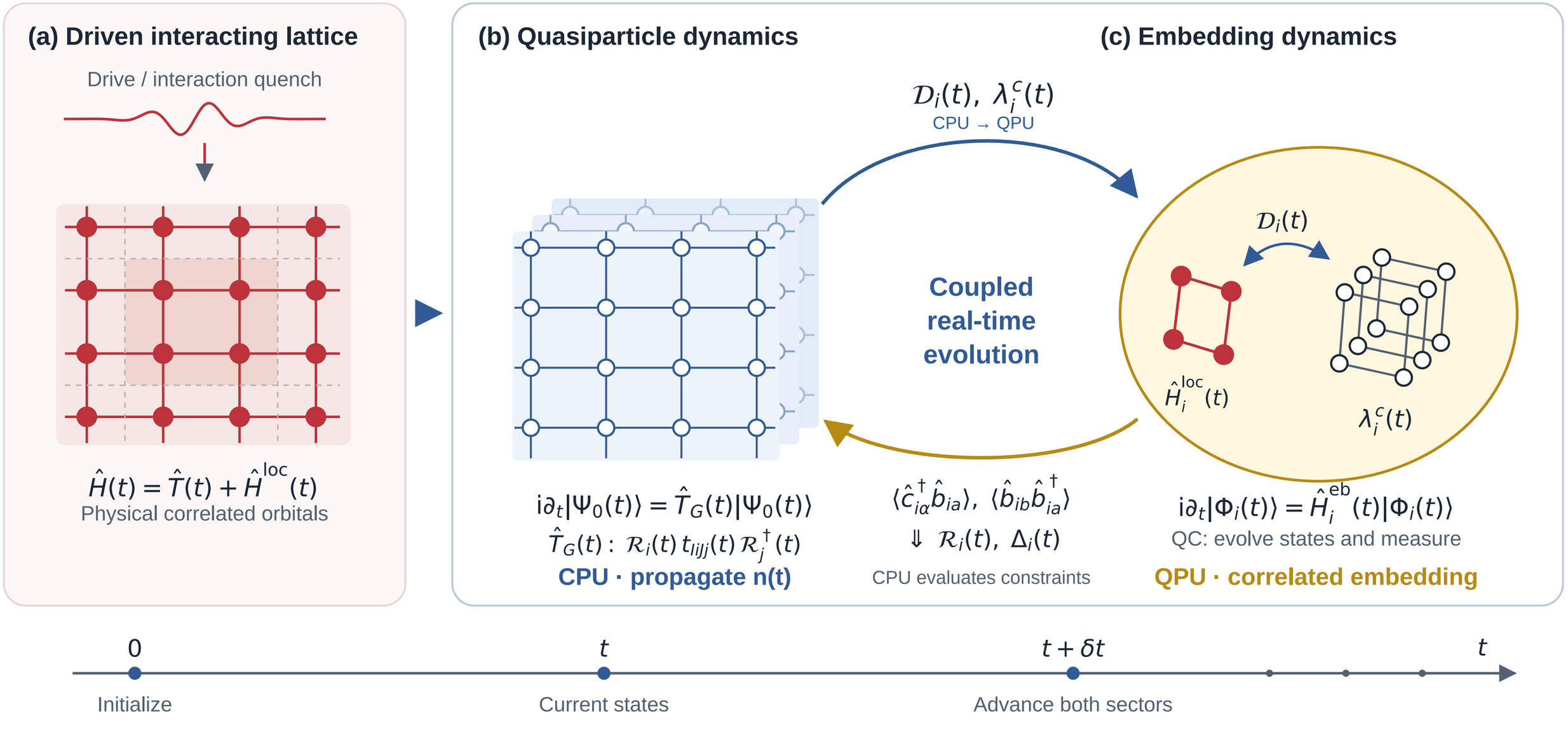

(a) Driven interacting lattice
Drive / interaction quench
$\hat{H}(t) = \hat{T}(t) + \hat{H}^{\mathrm{loc}}(t)$
Physical correlated orbitals
(b) Quasiparticle dynamics
(c) Embedding dynamics
$\mathcal{D}_i(t),\ \lambda_i^c(t)$
CPU → QPU
Coupled real-time evolution
$\mathcal{D}_i(t)$
$\hat{H}_i^{\mathrm{loc}}(t)$
$\lambda_i^c(t)$
$\mathrm{i}\partial_t|\Psi_0(t)\rangle = \hat{T}_G(t)|\Psi_0(t)\rangle$
$\hat{T}_G(t) :\ \mathcal{R}_i(t)\, t_{lijj}(t)\, \mathcal{R}_j^\dagger(t)$
CPU · propagate n(t)
$\langle \hat{c}_{i\alpha}^\dagger \hat{b}_{ia} \rangle,\ \langle \hat{b}_{ib} \hat{b}_{ia}^\dagger \rangle$
$\Downarrow\ \mathcal{R}_i(t),\ \Delta_i(t)$
CPU evaluates constraints
$\mathrm{i}\partial_t|\Phi_i(t)\rangle = \hat{H}_i^{\mathrm{eb}}(t)|\Phi_i(t)\rangle$
QC: evolve states and measure
QPU · correlated embedding
0
t
$t + \delta t$
t
Initialize
Current states
Advance both sectors